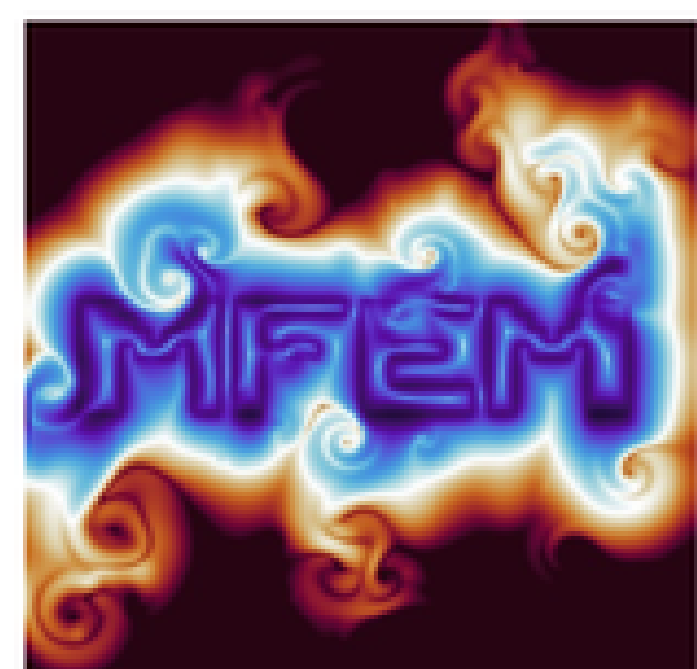
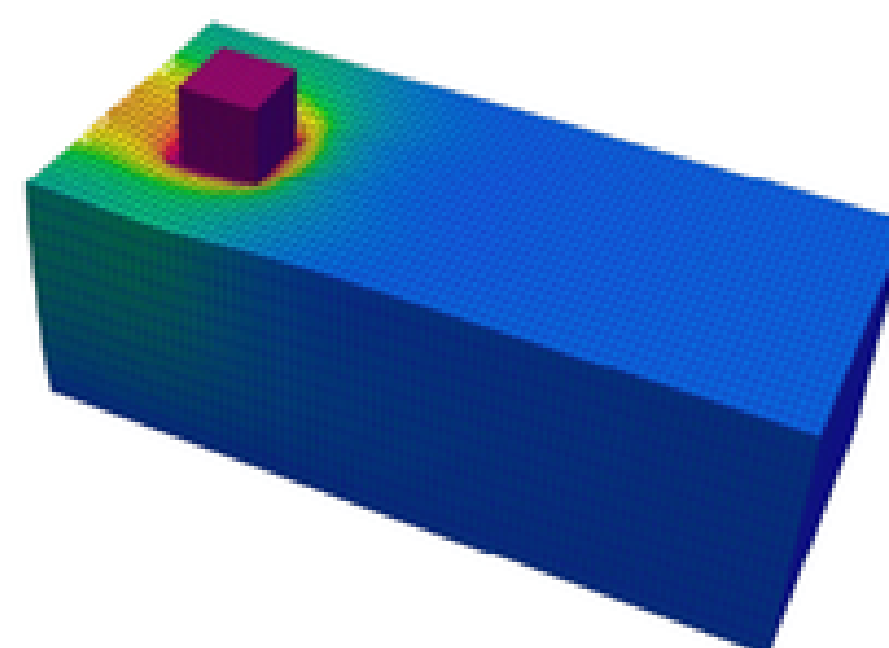


MFEM Introduction

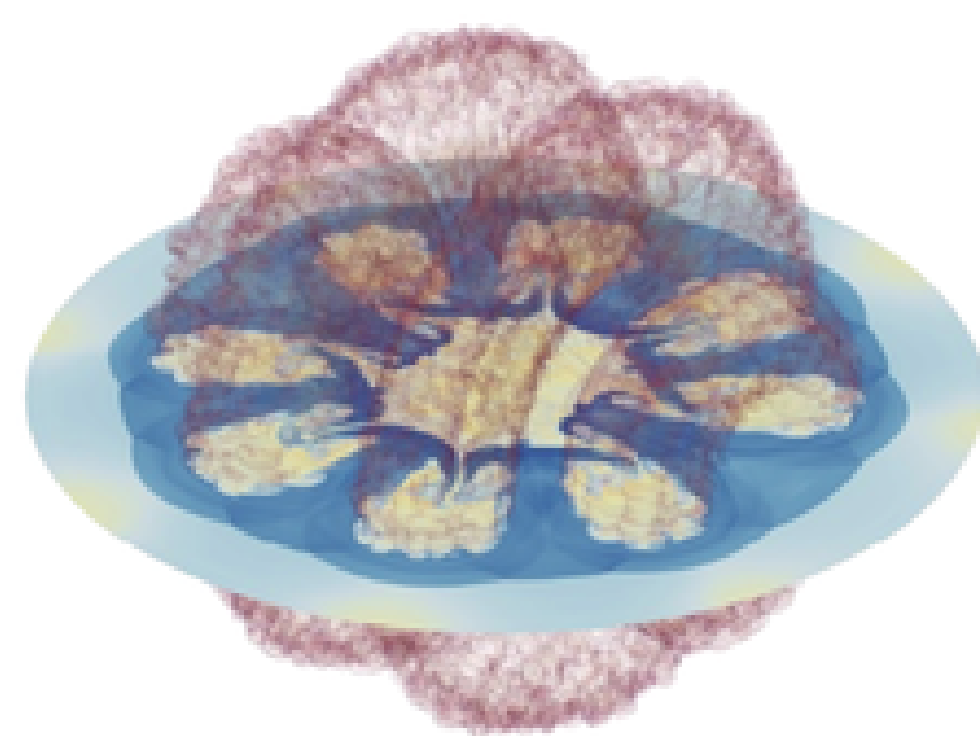
MFEM is a scalable, portable finite element package that provides the building blocks for developing finite element algorithms.



HDG convection-diffusion



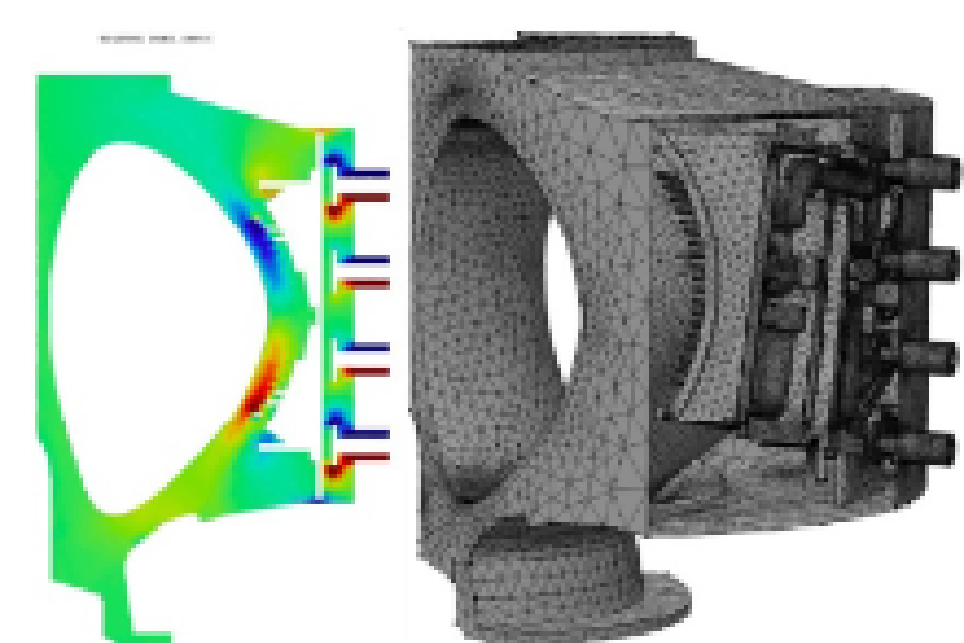
Contact mechanics



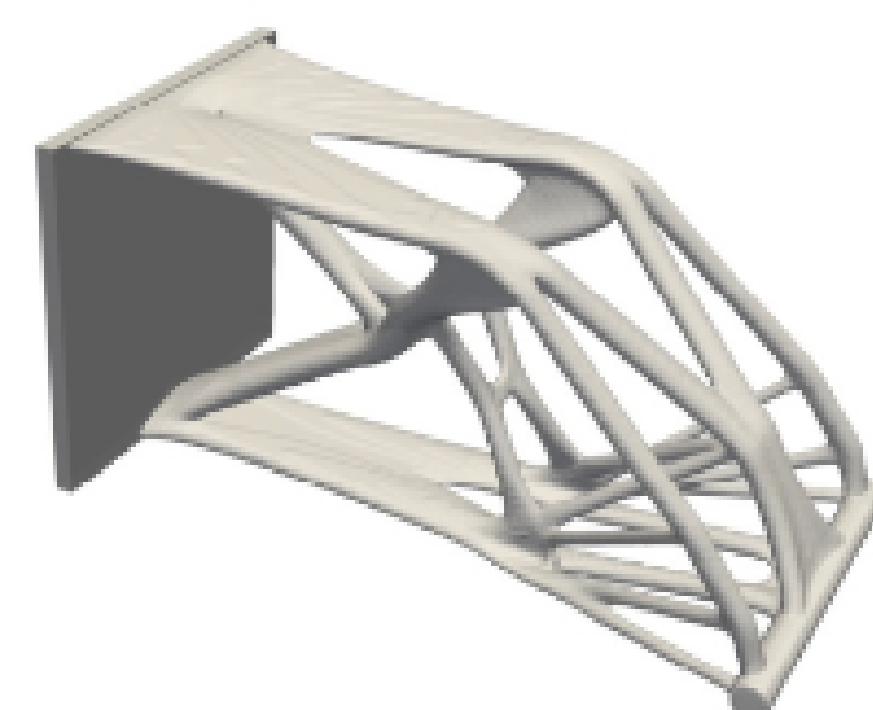
Compressible flow



Next-gen MRI



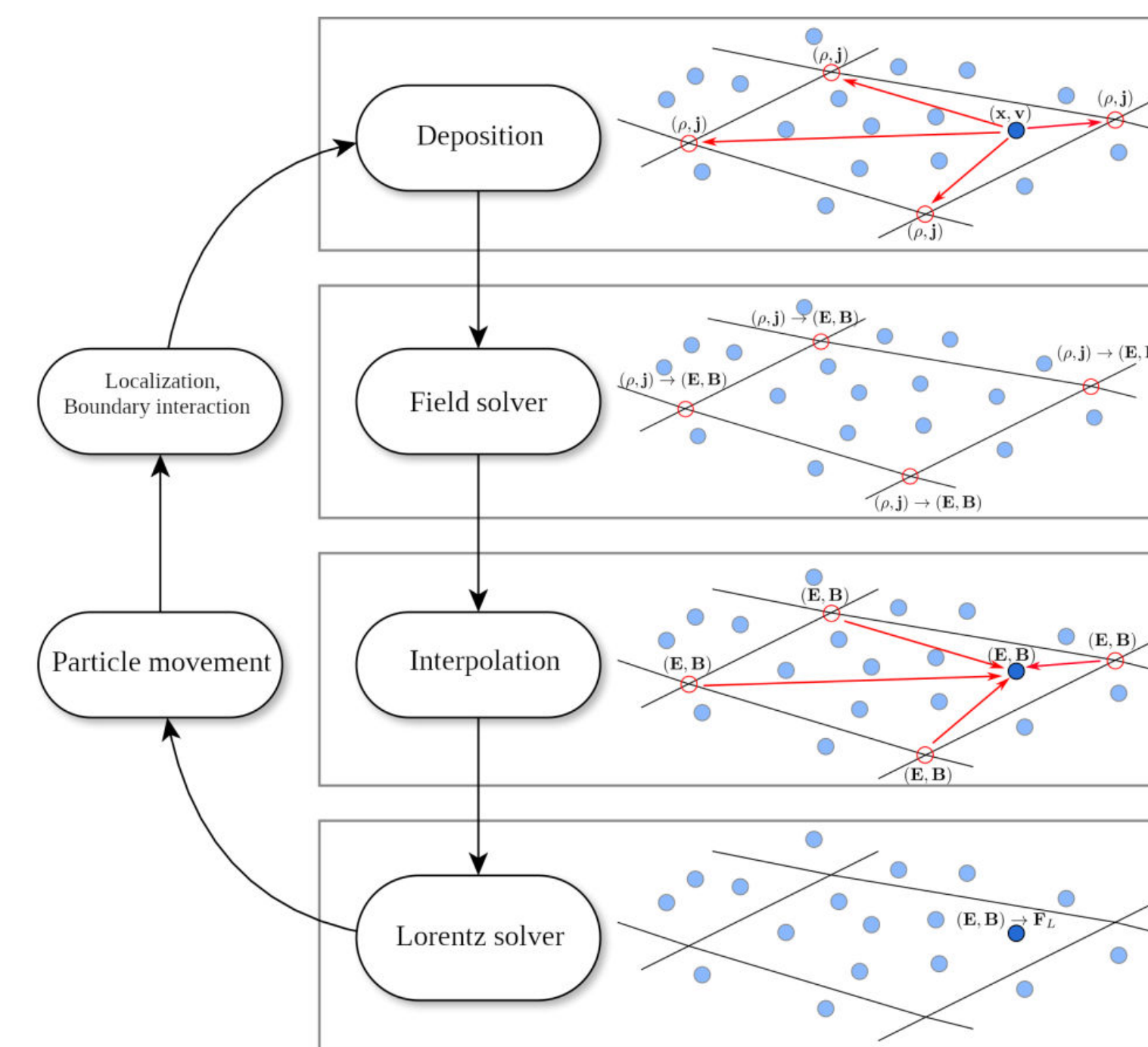
Core-edge tokamak



Topology Optimization

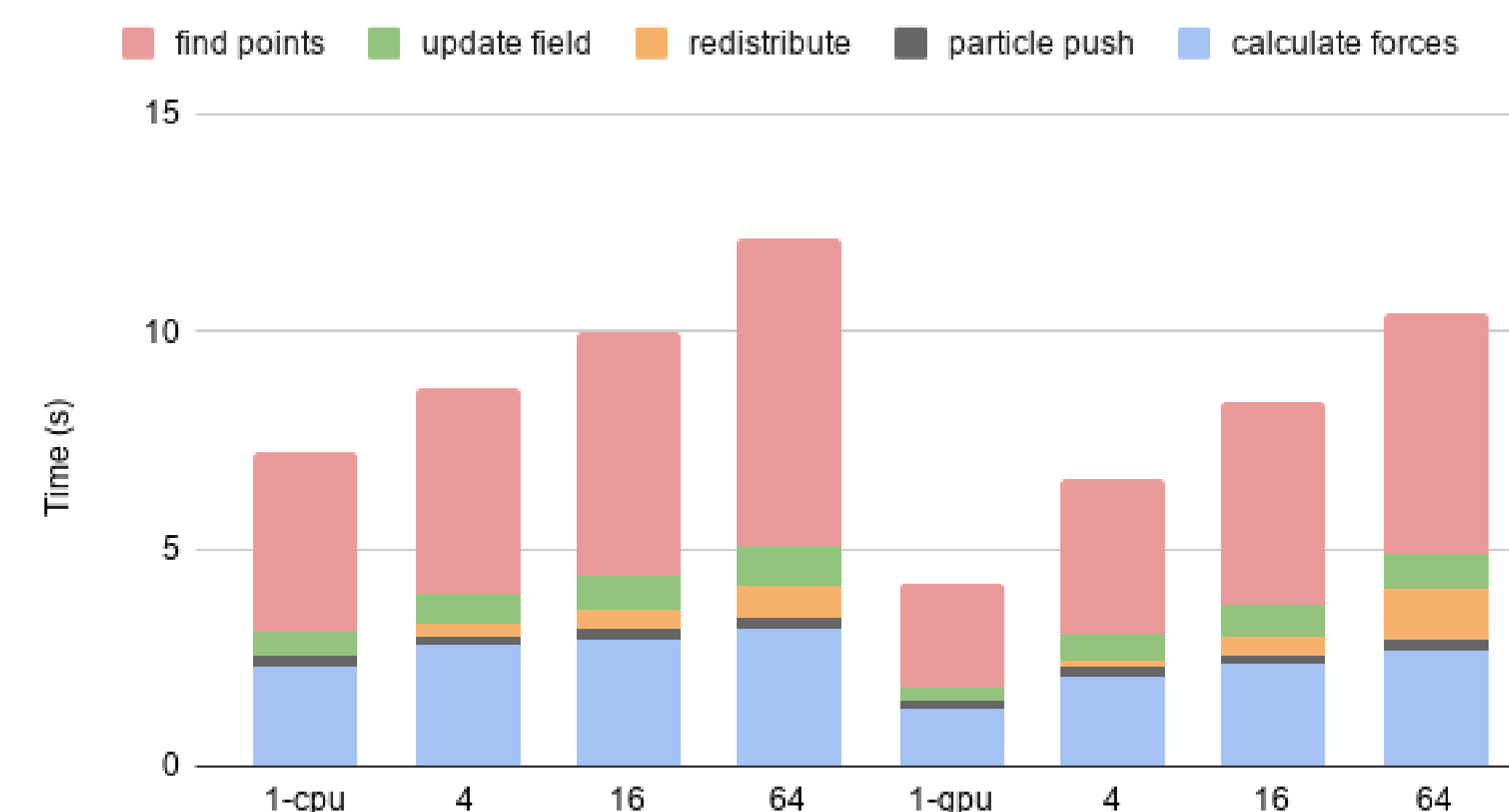
- Flexible discretizations on unstructured grids. Adaptive mesh refinement, mesh optimization
- High-order and scalable : MPI scalable to millions of cores and supports GPU-accelerated.
- Built-in solvers and visualization. **hypr**, **SUNDIALS**, **PETSc**, **SLEPc**, **SUPERLU** etc.

Part I: GPU Support for PIC Miniapp (Y. Luo)

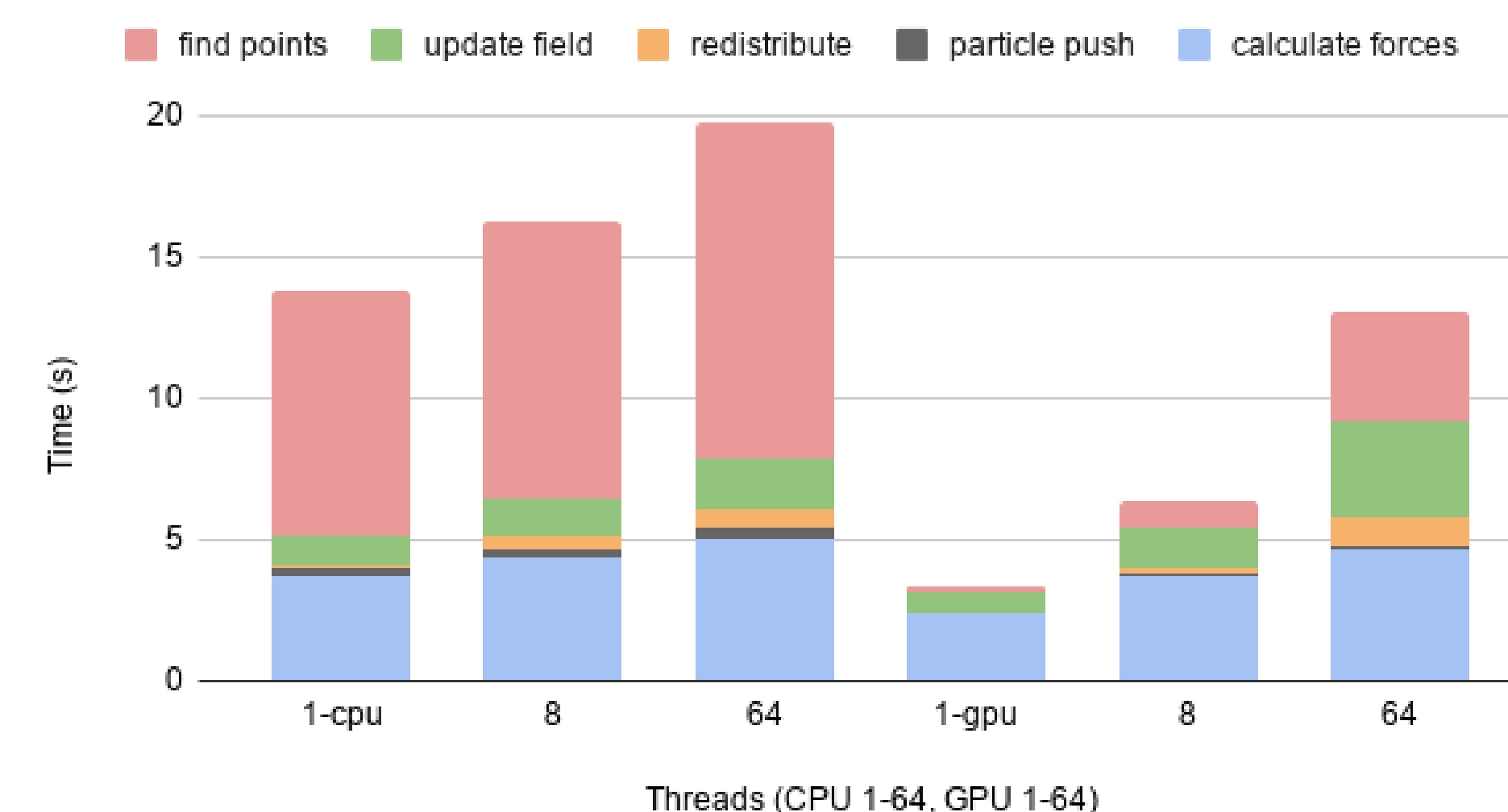


- Weak scaling results on 4 nodes with one A100 on each node from 1-64 threads.
- Note **FindPoints()** is called once in particle push and once in redistribute.

2D Mesh Particle In-Cell Time (10 Steps Total)



3D Mesh Particle In-Cell Time (10 Steps Total)



Part II: Crystal Router (E. Day)

- All-to-All communication algorithm to send particle data via MPI communications
- Repeatedly bisects data to be sent and kept, distributing particles across all ranks in $\log_2$ steps
- Modified to route an arbitrary amount of parameters and support routing of **MFEM** datatypes

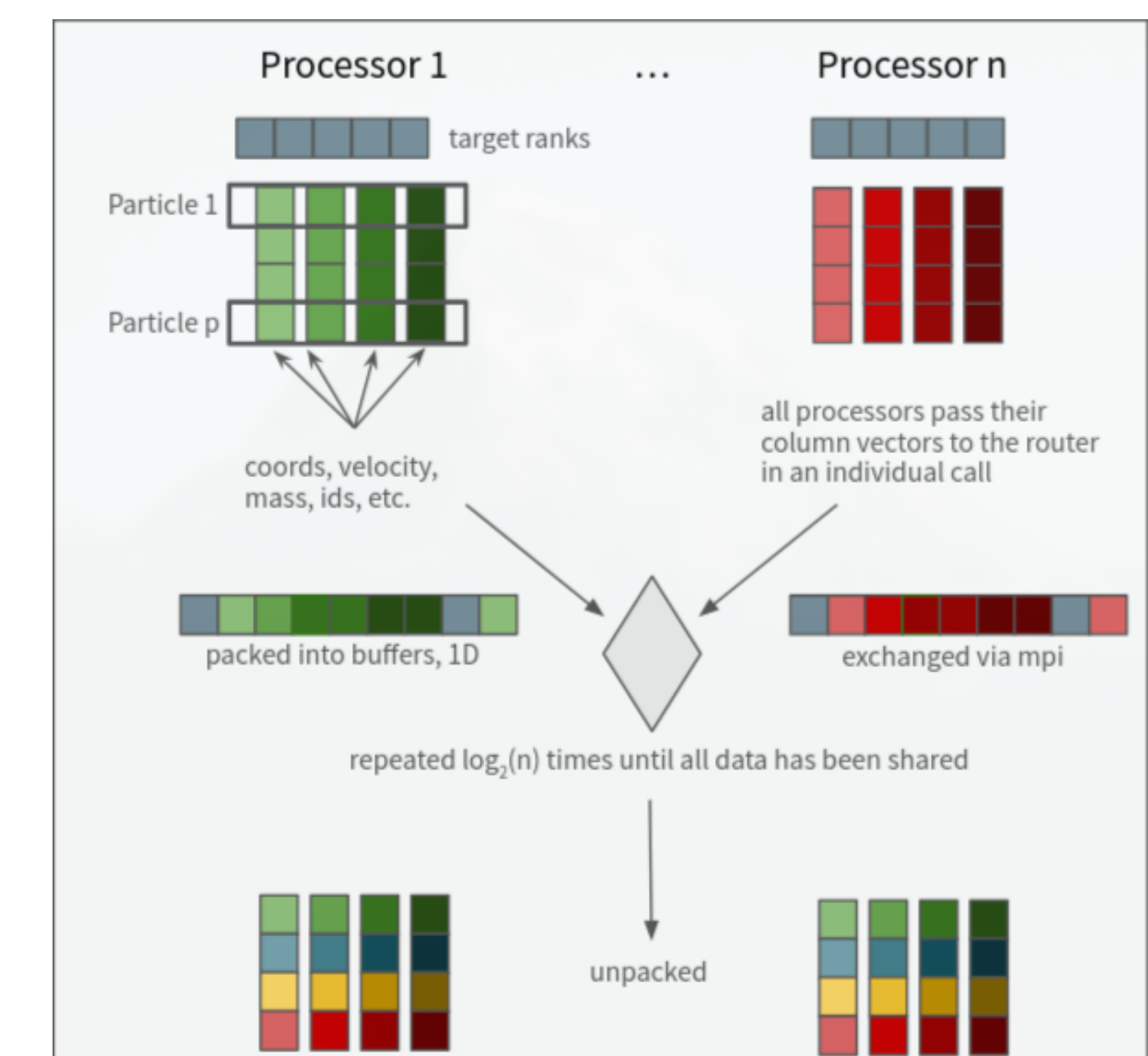


Figure 2. Example of how column vectors are packed within the router with target processor ranks and redistributed among processors for an all-to-all communication

New development during the intern:

- Miniapp to demonstrate particle redistribution functionality, working on 1,000,000 particles
- Unit tests to verify correctness of particles sent and destination

